Curriculum Vitae

Stuart Williamson
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1 Contact Information

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2 Research Experience

2.1 Generalized Predictions for the Electromagnetic Signatures of Mirror Stars

Undergraduate Researcher, University of Toronto — January 2024 - Present
Supervisors: Prof. Chris Matzner, Prof. David Curtin

- Characterization of mirror stars, a consequence of atomic dark matter, with the goal of predicting observables.
- From-scratch development of a python code to simulate and analyze mirror star hydrodynamics by numerical solution of stellar structure ODEs across a 3D parameter space.
- Implementation of various numerical techniques such as adaptive-step-size differential equation integration, multidimensional interpolation, contour fitting, root finding, and large dataset manipulation.
- `NumPy`, `SciPy`, `Matplotlib` used to efficiently create and display data.
- Code available on Github. Paper under review with preprint available on arXiv. Further details on project available on my personal website.

2.2 Trans-correlated hamiltonians in Q-SENSE

Summer Research Student, University of Toronto — May 2025 - October 2025
Supervisor: Prof. Artur Izmaylov

- Working in the second quantization formalism of quantum mechanics to develop an efficient method for solving the electronic structure problem.
- Adapting the group's hardware efficient Q-SENSE method to work with trans-correlated hamiltonians, presenting challenges of 3-body operator terms and non-hermiticity.
- Linear algebra-heavy numerical work with computational efficiency a priority: Large sparse matrix manipulation, high-dimensional tensors, subspace matrix diagonalization and construction.
- Python packages learned: Google's `Openfermion`, the open-source project `ffsim`, alongside `NumPy`, `SciPy`, and `Matplotlib`.
- Use of SSH to control remote hardware for large calculations.

2.3 Gravitational Wave Polarizations in Next-Gen Detectors

Undergraduate Researcher, University of Toronto — September 2025 - April 2026
Supervisor: Prof. Reed Essick

- Investigating the capability of next-generation detectors to distinguish the up to six possible polarizations of gravitational waves in general metric theories of gravity.
- Results will support next-generation detectors' science case and inform the ideal location of their placement.
- Simulation of a detector network's response to astrophysical signals as a function of signal source location, arrival time, signal shape, signal polarization, detector count, detector location, and detector size.
- Matched filtering used to quantify a detector network's ability to identify or distinguish signal parameters.
- Fourier-domain signal creation, analysis and manipulation.
- Use of, and modification to, Prof. Essick's `GW-Detectors` package along with `NumPy`.
- Project report available here.

3 Academic History

University of Toronto H.B.Sc. June 2026

Astronomy & Physics Specialist, Mathematics Major — GPA: 3.98/4.00

Notable Activities

AST425
Research topic in Astronomy: 3 class presentations, LaTeX typeset project proposal and final report. See section 2.3 for project description.

AST325
Three LaTeX typeset lab reports on astrophysical concepts tested in lab: Photon statistics with CCDs, Fourier analysis of thermal radio signals for calibration, Spectrographic calibration of 1D and 2D spectrometers.

AST221
15 page LaTeX typeset report on colour magnitude diagrams of 3 GAIA clusters: Distance determination and isochrone fitting to determine cluster age, including cluster membership selection by proper motion and parallax.

UTAT Space Systems
Engineering design team creating FINCH, a 3U cubesat to be launched into orbit 2025. Personally worked on the Attitude Determination and Control System, simulating and analyzing orbital paths and satellite orientation to present findings to other systems.

4 Professional Experience

Student Library Assistant September 2024 - May 2025
UTL Collection Development: *130 St. George St. Toronto, ON*

- Received media from various international companies, processing invoices for department review and sorting items for addition to a number of university library collections.
- Physically processed items with barcodes and anti-theft measures based on item type.
- After cataloguing, applied item identifier and used library software Alma to mark item for distribution.

Student Library Assistant September 2023 - August 2024
UTL at Downsview: 4961 *Dufferin St, North York, ON*

- Processed books sent from other libraries into the facility storage standard, applying unique barcodes and updating relevant database information.
- Handled books and other forms of media with the care and attention appropriate for long-term preservation of delicate objects.

Assistant Bookkeeper May - August 2023
UWC Bookkeeping Services: *Innisfil, ON*

- Receipts, invoices, bills and statements from the client were confirmed accurate and entered into the clients database. These were then sorted and filed neatly.
- Confirmed and reconciled bank and credit card statements.
- Organized clients digital file systems for easier record keeping in the future.

5 Awards & Scholarships

Later Life Learning OSOTF Award	$400	2023, 2024
Innis College Alumni And Friends Award	$400	2025
George Luste (In Program) Scholarship	$3,000	2024
John Pounder Prize In Astronomy	$1,900	2024
The H. S. Robertson Prize In Astronomy	$1,350, $1,990	2024, 2025
Walter John Helm Scholarship in Astronomy	$2,070	2025
Boris Stoicheff Memorial Scholarship in Physics	$2,900	2025
CQIQC Summer Undergraduate Research Award	$10,000	2025